

Clinton Judy

Boalsburg, PA | clinton@judy.io | github.com/judy | linkedin.com/in/clintonjudy | clinton.judy.io

SUMMARY

I build production AI products end to end. At Zipline I created Zippy, the company's flagship AI assistant: a retrieval system answering frontline retail workers across 50+ brands and 1,000,000+ conversations, which I then re-architected twice underneath live traffic and extended with agentic tool use over MCP. I shipped contextual, atomic-fact retrieval to paying customers about fifteen months before Anthropic published the same core idea as Contextual Retrieval. That work rests on 15+ years as a Rails engineer and SRE operating at scale, and today I'm building a from-scratch agent runtime with its own memory and emotional substrate.

WORK EXPERIENCE

Senior Web Engineer

May 2019 — Sep 2026

Zipline

Remote

AI Systems (2023–2026)

- **Created Zippy, Zipline's production RAG assistant and flagship AI feature:** found the approach in research, pitched it, built the prototype in two weeks, named it, and shipped it. Live with **50+ retail brands** across **1,000,000+ conversations** and **2,000,000+ pages** indexed; the product's primary AI differentiator for three years.
- Arrived independently at contextual retrieval about fifteen months before it was published as a technique: built ingestion that decomposes documents into self-contained **atomic facts**, the same core insight Anthropic later published as Contextual Retrieval.
- **Re-architected the retrieval engine underneath a live product, twice.** Led the migration from pgvector to a hybrid Elasticsearch index (77-ticket program, 97% delivered), then designed and shipped vision-model document understanding so content that exists only as diagrams, planograms, and tables became answerable.
- **Shipped agentic tool use to production**, including MCP (Model Context Protocol) integration that lets Zippy query live systems for real-time inventory, scheduling, and policy answers. Built the MCP server model, client, and admin interface, and multi-step tool calling via the OpenAI Responses API.
- **Made retrieval quality measurable:** co-built Zippy's first LLM-as-judge evaluation system and the evaluation corpora later architecture decisions were judged against, including a scenario set generated from three years of real customer bug reports, so migrations were decided on measured pass rates instead of opinion.
- Engineered fast, access-controlled retrieval at enterprise scale: vector search combined with permission-predicate filtering, working around HNSW's degradation under predicate constraints so results stay fast and correctly scoped per user, team, and role.
- Built the full conversational front end with Hotwire (Turbo + Stimulus): token-streaming responses, multi-turn state, source citations, query refinement, and inline feedback capture.
- Shipped a multilingual retrieval fix over a weekend to unblock a customer's pre-recorded national television segment, adding automatic query translation so non-English questions retrieved correctly.

Architecture & Technical Leadership (2025–2026)

- Settled architecture arguments with measured evidence: ran a corpus study on real customer documents to prove which extraction failures actually occurred, won committee approval for the result over the chair's preferred alternative, and had it in production fifteen working days later.
- Set the numeric go/no-go criteria my own projects were judged against, before results came in.
- Challenged a projected \$100K/yr infrastructure spend by pricing it against the existing pipeline's invoiced cost, then ran the vendor review that surfaced a 32x cheaper path.
- Served as the company's AI technical authority: standing member of the AI Steering Committee, author of the RFCs setting AI architecture direction, escalation point for AI questions from engineering, product, sales, and customers; fronted the platform's AI story in enterprise security and responsible-AI reviews, sales RFPs, and vendor negotiations.
- Acted as team lead for the AI/search team: cycle planning, technical direction, and distributing architecture work across the team rather than holding it.

Product & Platform Engineering (2019–2023)

- Built Assessment tools end to end: historic results, auditee search, scoring and sorting, and the complete auditor workflow. **116M+ assessment questions answered.**

- Built Report Builder (custom columns, filters, role-based access), Learning Insights learner reports, and discussion threading with notifications and mentions.
- Built platform capabilities: REST Hooks for Zapier, a video transcoding pipeline on AWS MediaConvert with HLS streaming, and MegaSeed, an internal large-scale realistic-data generator for performance testing.
- Migrated legacy reporting to BigQuery as part of a small team, matching edge cases exactly so historical numbers stayed correct.
- Kept the application fast as Zipline grew 10x in active users, through Rails and SQL optimization, caching, and automated performance testing in CI.

Software Engineer

Oct 2018 — Apr 2019

Flatiron School

New York, NY

- Consolidated logs, metrics, and traces across a large microservice architecture, increasing visibility and reducing MTTR; helped migrate services from Digital Ocean to AWS.

Senior Site Reliability Engineer

Dec 2016 — Oct 2018

AccuWeather

State College, PA

- Built the CI/CD pipeline and a monitoring system tracking hundreds of statistics across tens of millions of daily requests; automated deploys across public-facing AccuWeather properties. Contributed to Ansible and Terraform.

Senior Web Developer

Feb 2016 — Oct 2016

EFC Systems

State College, PA

- Built data-visualization dashboards (JavaScript + D3) and front-end tooling for precision agriculture; overhauled authentication to modern encryption standards.

Senior Rails Engineer

Nov 2013 — Jan 2016

Agrian Inc.

State College, PA

- Built service-oriented Rails applications and large OO frameworks for diverse raw data formats, optimized for big-data scale (100M rows in the first month).

Open Source Lead

2012 — 2013

Fractured Atlas

Remote

- Turned Artful.ly, a CRM and online ticketing platform, into an open-source Rails engine; built discount, search, and customer-segmentation features and improved sign-up conversion.

Web Developer

2009 — 2011

West Arete

State College, PA

- Designed, built, and launched data-driven Rails web applications for clients including MIT, Penn State, the DOT, and the Marine Biological Laboratory.

Founder & Community Organizer

2010 — 2019

The Make Space / State College Ruby Group

State College, PA

- Founded The Make Space, a State College hackerspace, and ran the State College Ruby Group for several years, giving a recurring series of technical talks; consulted local startups on architecture, UX, and object-oriented design.

SELECTED PROJECTS

Kormos, Plektos & Thymos: An Agent Runtime with Memory and Emotion

Mar 2026 — Present

Independent

- Building Kormos, a from-scratch agent runtime (Rails 8) that hosts multiple LLM agents with tool calling, scheduled autonomy, and a first-class MCP client layer (OAuth via Dynamic Client Registration, per-provider token lifecycle).
- Building Plektos, an episodic + semantic memory system with a dual-embedding retrieval model (what happened plus what it felt like), consolidation, and salience-weighted recall: the organizational-memory problem applied to a single agent.
- Built Thymos, an emotional-signal substrate distilled from a broad survey of affective-science research into one coherent model: hormone-like signals (cortisol, dopamine, and others) with realistic temporal dynamics and homeostatic resting points that compose into internal state, giving agents a basis to resist sycophancy and disagree rather than reflexively agree.
- Run a custom agent-driven coding pipeline with humans in the loop at strategic decision points and dedicated automated architectural-review stages, keeping code cleanliness and structural quality a first-class priority as the system grows.

SKILLS

- **AI/ML:** Retrieval-Augmented Generation (RAG), contextual retrieval, Large Language Models (LLM), OpenAI Responses API, embeddings, vector search, agentic systems, multi-agent orchestration, episodic memory / consolidation, MCP (Model Context Protocol) client design with OAuth Dynamic Client Registration
- **Languages & Frameworks:** Ruby on Rails, Hotwire (Turbo + Stimulus), JavaScript; Svelte for reactive frontends; React (in production at Zipline)
- **Data:** PostgreSQL, pgvector, Elasticsearch (incl. Painless scripting), Redis, BigQuery, Sidekiq, Action Cable
- **Infrastructure:** Docker, Ansible, AWS, Terraform, CI/CD
- **Other:** D3, video transcoding (HLS), open-source contributions (PDF parsing, Ruby gems)

EDUCATION

Lock Haven University of Pennsylvania

Lock Haven, PA

Coursework in Computer Science, Graphic & Online Design

Aug 2004 — May 2007